

Deep Inelastic Scattering Revisited

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Abstract

Discuss Yadism [\[1\]](#).

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Chapter 1

The basics

1.1 Introduction

1.1.1 Motivation and non-goals

We aim in this course to make a meaningful comparison between the theoretical predictions and the experimental measurement of a fully inclusive Deeply Inelastic Scattering (DIS) cross-section. Instead of an abstract, theoretical discussion about DIS, which can be found in a standard lecture on particle physics, we're looking here to make a connection with real-life physics solving current research questions. The main focus is be naturally on PDF, for which DIS plays a major role. In fact, DIS was, is, and will be a very important constrain in determining Parton Distribution Functions (PDFs) - only recently, e.g. starting from NNPDF4.0 [2], LHC data starts to dominate in PDF fits. Stressing the importance and relevance of PDFs for any kind of perturbative QCD (pQCD) application and to understand the inner structure of hadrons is left to the literature [3–5]. In turn, DIS itself can be used to answer unresolved questions about PDFs, e.g. whether the proton has an intrinsic charm component [6].

In order to obtain a more realistic DIS treatment, we review general pQCD features and exemplify them using DIS. In fact, fully inclusive DIS comes in very handy, since here the effects mostly appear in their simplest form and when considering hadronic collisions, e.g. at the LHC, the strategies are mostly the same just more involved. We keep a phenomenological point of view with a specific focus on the numerical implementation as, eventually, we want to compare to the experimental measurements. We refrain from giving detailed proofs in this course and point to the standard textbooks [7–11] for that.

Instead, we consider different features as given and focus on their practical implications. However, a major complication in this course is the interconnection between almost all features and while we discuss them here in turn, we highlight also how they are intertwined with each other.

In recent years (and much more so hopefully in the coming years), DIS has become a major research topic again thanks to the advent of the Electron Ion Collider (EIC) [12]. While the planned physics program of the EIC goes far beyond standard fully inclusive DIS, this simplest case can serve as an introduction into more complicated topics.

The structure of this course follows in large parts the implementation of Yadism (Yet Another DIS module) [1, 13], which is developed and used by the NNPDF collaboration in their PDF determinations. The online documentation, which is available at

<https://yadism.readthedocs.io/> ,

may serve as additional, more technical introduction into the topic. Note that Yadism is an open-source project to which everybody can contribute¹.

Since DIS is a very wide field, it is worth spelling out explicitly some non-goals:

- We only consider pQCD in the collinear framework here. However, because DIS is such a powerful experiment, which offers a clean access to the inner structure of the protons it can be studied using other frameworks as well, such as Color Glass Condensates [14, 15] → ask H. Mäntysaari or T. Lappi
- We only consider fully inclusive cross-sections here (to be defined in Section 1.2 and with a tiny exception to be discussed in Section 2.5). In particular, we do not consider any more complicated final states, such as, e.g., single jet or di-jet production [16].
- We only consider pQCD in the collinear framework here. While, Monte Carlo Event generators, such as, e.g., Pythia [17], are rooted in collinear pQCD they eventually go much beyond and we do not consider their details here [18] → ask I. Helenius.
- We only consider DIS here. In particular, we do not consider photo-production or elastic or diffractive cross sections. We define the necessary kinematics for the distinction below in Section 1.1.3.

¹e.g. if some documentation can be improved

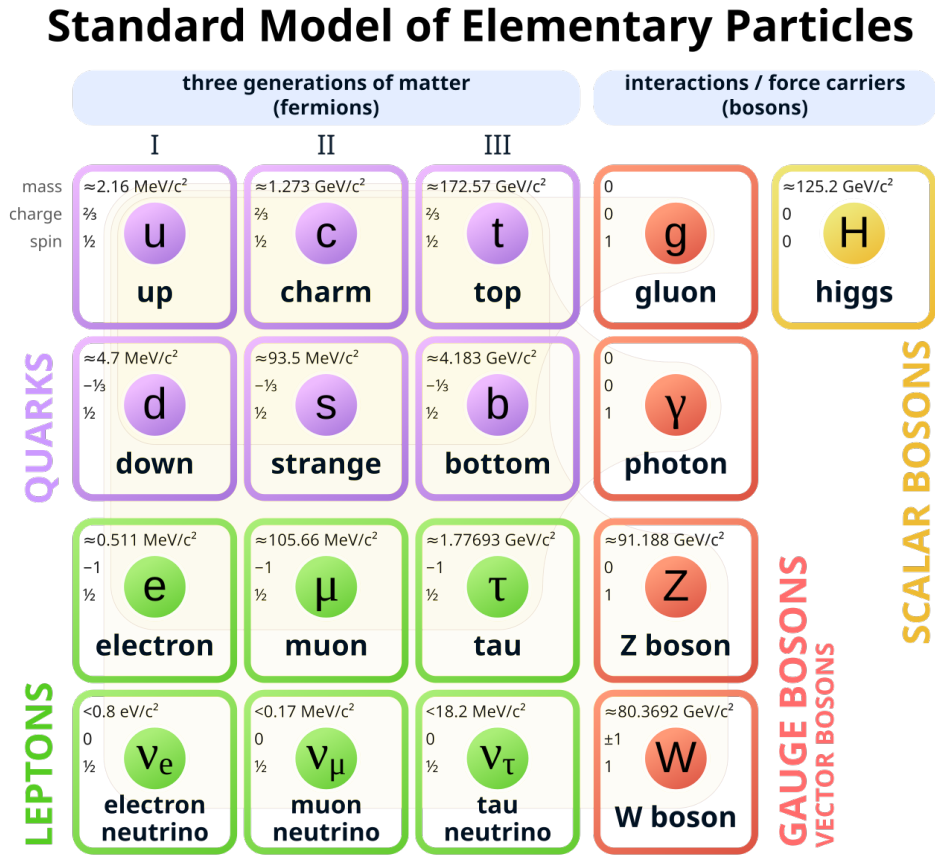


Figure 1.1: The Standard Model of Elementary Particles - taken from [19]

- We compute the theoretical predictions here. So while we try to keep the connection to the experimental side close, we do not discuss any specific details about how to measure cross-sections.

1.1.2 Requirements

We need some basic knowledge from particle physics and, in particular, you need to be familiar with the Standard Model overview shown in Fig. 1.1. We discuss in this course each and everyone of these particles, with the exception of the three heaviest in each category: the heaviest quark, the top, the heaviest lepton, the tau, and the heaviest boson, the Higgs. They work, in principle, mostly like their lighter brothers and sisters, but in practice are basically always neglected. We give a more thorough justification to discard them when we get to the actual numbers. All other particles explicitly appear in this course and you need to know their charges - see also Fig. 1.1.

Summary forces, particles, and charges

- The carrier of the strong force is the gluon
- Quarks have a strong charge, leptons don't
- The carriers of the electro-weak force are the photon, the Z boson, and the W boson
- Quarks and charged leptons have electro-magnetic charges
- Quarks and leptons have weak charges

We also need some basic knowledge from Quantum Field Theory.

1.1.3 The very basics

Let's start from the top:

DIS in words

fully inclusive DIS = fully inclusive deeply inelastic scattering

What does that actually mean? We begin from the back:

- “scattering” means, of course, we are scattering two particles with each other. In this particular case, we are scattering a lepton off a hadron and to make things simple in the beginning we consider an electron is scattered off a proton.
- “inelastic” means we are breaking the target, the proton, apart; it shatters into many pieces.
- “deeply” is an adjective² to inelastic and it means we are looking “deep” into the inside of the proton. And in order to resolve small things, as usual, we need an highly energetic probe.
- “fully inclusive” means we are observing a single particle in the final state, which is the electron, but nothing else.

Thus, we can write the same information again with a formula

²thus it is “deeply inelastic” or “deep-inelastic”, but not a list of adjectives “deep inelastic”

DIS in a formula

fully inclusive DIS = $e^-(k) + p(P) \rightarrow e^-(k') + X$ with $-(k - k')^2 \gg \Lambda_{\text{QCD}}^2$

where we have assigned the four-momenta k, P, k' to the incoming electron, the incoming proton and the outgoing electron respectively. As usual X marks the inclusiveness, i.e. we don't observe anything else apart from the electron. By requiring a large momentum-transfer between the incoming and the outgoing electron, i.e. their difference four-momentum vector has a large magnitude, we probe the proton with highly energetic parton (which we can think of as photon).

Next, we need a way to actually compute theory predictions for the cross-sections, which you can measure in an experiment. We use the factorization theorem [20],

Factorization Theorem

$$\sigma(x, Q^2) = \sum_j \int_x^1 \frac{dz}{z} C_j(z, Q^2) f_j(x/z, Q^2) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \quad (1.1)$$

$$= \sum_j (C_j(Q^2) \otimes f_j(Q^2))(x) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \quad (1.2)$$

$$= (\mathbf{C}(Q^2) \otimes \mathbf{f}(Q^2))(x) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right), \quad (1.3)$$

which states how to compute a cross-section $\sigma(x, Q^2)$ from coefficient function $\mathbf{C}(z, Q^2)$, computable in pQCD, and a universal PDF $\mathbf{f}(y, Q^2)$. In the remaining part of the course we discuss, enhance and correct this formula in many different ways and basically challenge every aspect. We define each and every variable and function in Section 1.2, so we can focus for the moment on two important aspects: universality and notation.

The crucial, physical aspect about Eq. (1.1) is universality: for any cross-section σ measured at any x and Q^2 there is exactly one PDF $\mathbf{f}(y, Q^2)$. Instead, the coefficient functions $\mathbf{C}(z, Q^2)$ may depend on the definition of the cross-section, but they are computable from first principles. This implies the predictive powers of our framework: if $\mathbf{C}$ is fixed and $\mathbf{f}$ is universal, we can measure σ in one set of experiments and then predict the outcome of another set of measurements σ' .

The other, more practical aspect of Eqs. (1.1) to (1.3) is the short-hand notation we introduce there. While Eq. (1.1) is the most canonical form of the factorization theorem, which can be found like this in most textbooks, we introduce in Eq. (1.2) the multiplicative convolution symbol $\otimes$. Note that we have suppressed the variable over which the convolution is performed and recall that convolution is associative, i.e. $(g \otimes h)(x) = (h \otimes g)(x)$. Finally, in Eq. (1.3) we introduce the vector notation, where we denote object, which have a non-trivial dependence on the parton flavor j , with bold font. Equation (1.3) looks nice and compact, but it worth recalling that it already hides a lot of details: specifically, $C_j(z, Q^2)$ is an object which depends on three different variables.

- LO

1.2 Setting the stage

- x, Q^2
- $L_{\mu\nu} W^{\mu\nu}$

1.3 Distributions

As we have seen the coefficient functions may contain distributions. For example the quark coefficient function at LO is just a Dirac delta distribution:

$$C_q^{(0)}(z) = \delta(1 - z). \quad (1.4)$$

To better compare with our next step, let's repeat the definition of the Dirac delta distribution:

$$\int_0^1 dz \delta(1 - z) f(z) = f(1). \quad (1.5)$$

However, this is only the simplest case, which appears at LO, and at higher order we get more complicated distributions. For example the quark coefficient function at NLO is

given by

$$C_q^{(1)}(z) = C_F \left[4 \left(\frac{\ln(1-z)}{1-z} \right)_+ - 3 \left(\frac{1}{1-z} \right)_+ - (9 + 4\zeta_2) \delta(1-z) \right. \\ \left. - 2(1+z) \ln \left(\frac{1-z}{z} \right) - 4 \frac{\ln(z)}{1-z} + 6 + 4z \right] \quad (1.6)$$

where the first line only contains distributions. Here, we have introduced the usual $+$ -distributions, which are defined via

$$\int_0^1 dz a(z)_+ f(z) = \int_0^1 dz a(z) (f(z) - f(1)) . \quad (1.7)$$

We review in Section 1.4 again briefly on how these distributions appear, but for the moment we turn to more practical matters as the goal of our course is to get an actual comparison to experimental data. The problem is: distributions are (complicate) math objects, defined only via integral relations, Eqs. (1.5) and (1.7), which are unsuitable for numeric evaluation.

1.4 Next-to-leading order

- NLO_q
- NLO_g

We need the Riemann Zeta function $\zeta(n)$ below, which is defined at integer values n by

$$\zeta_n = \zeta(n) = \sum_{j=1}^{\infty} \frac{1}{j^n} \quad (1.8)$$

Chapter 2

DIS in real life

2.1 DIS is diverse

- γ/Z
- leptons
- hadrons: p+d+A; nucl. PDF
- experimental overview

2.2 Interpolation grids

2.3 DGLAP

- resummation
- Mellin

2.4 Scale variations

- scheme A/B/C
- MHOU

- covariance matrix

2.5 Heavy Quarks

- F_2^c
- FONLL

2.6 γ_5

- helicity
- HVBM + Larin

2.7 TMC

Chapter 3

More advanced topics

3.1 Positivity

- DIS vs. MSbar

3.2 Higher orders

- IHOU
- N3LO
- channels
- loops and legs

3.3 QED

- Qiu et al.

3.4 Valo

- $C_\gamma \rightarrow k$

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